Where the Revenue Comes From: a UK Online Gift Retailer

Dec 2009 to Nov 2011 | 1,067,371 order lines | Excel workbook with live formulas | Prepared by Muhammad Sadat Samin

Question

Which customers, products, countries and months bring in the revenue, how much leaks out through returns, and what would I change if I ran this shop?

Data and cleaning

The source is *Online Retail II* from the UCI Machine Learning Repository (CC BY 4.0), which records every order line of one UK online giftware seller, in pounds. The analysis covers 24 full months, 1 Dec 2009 to 30 Nov 2011. The 25,526 lines from 1 to 9 Dec 2011 are dropped because that month is incomplete. The two raw sheets overlap for 1 to 9 Dec 2010 (22,523 identical lines), and that stretch is counted once. 170 product codes appear in both upper and lower case (85123A and 85123a) and are merged, which leaves 4,720 products. Invoices starting with C are cancellations and count as returns. Postage, bank fees, discounts and manual adjustments are not products, so they sit outside product revenue and are shown on their own (net £95,987). The 5,975 lines with a price of zero or less are excluded. 21.77% of lines have no customer ID. They stay in the revenue totals but not in the customer analysis.

Method

Line revenue is quantity times price, and net revenue is sales plus returns. An order is one sales invoice. Average order value (AOV) is gross sales divided by orders. Year 1 runs from Dec 2009 to Nov 2010 and Year 2 from Dec 2010 to Nov 2011. Customers get a 1 to 5 score for recency, frequency and spend (RFM) as of 1 Dec 2011, using quintile cut-offs, and fall into five segments. The forecast uses Excel's ETS function on 24 monthly points. Code built the invoice, product, customer and month-by-country tables, and every analysis sheet reads them through formulas. The period and country dropdowns on the dashboard drive its headline boxes.

	Year 1	Year 2	Change	24 months
Gross sales	£9,428,750	£9,655,545	+2.41%	£19,084,295
Returns	-£240,932	-£304,597	+26.42%	-£545,530
Net revenue	**£9,187,818**	**£9,350,947**	**+1.78%**	**£18,538,765**
Orders	19,743	18,957	-3.98%	38,700
Average order value	£477.57	£509.34	+6.65%	£493.13
Return rate	2.56%	3.15%	+0.60 pts	2.86%

Customer segment	Customers	Share	Net revenue	Revenue share	Rule
Champions	1,225	21.0%	£11,026,921	68.6%	R, F and M scores all 4 or 5
Loyal	1,310	22.5%	£2,511,646	15.6%	R and F 3 or more, not Champions
Potential	962	16.5%	£505,789	3.1%	R 3 or more, F under 3
At risk	1,386	23.8%	£1,648,520	10.3%	R = 2, or R = 1 with F 3 or more
Lost	941	16.2%	£381,328	2.4%	R = 1 and F under 3

Product lines only. The customer table covers 86.7% of net revenue, because lines without a customer ID cannot be assigned.

Findings

Net revenue rose 1.78%, from £9.19M to £9.35M, even though orders fell 3.98%. Baskets got bigger: AOV went from £477.57 to £509.34, up 6.65%.

September to November brought in £6.83M, which is 36.82% of the 24 months, and November was the biggest month in both years (£1.40M and £1.43M). December drops back (£781,379 and £760,258), but sales stop on 23 December in both years, so it has only 21 and 20 trading days. Pre-Christmas shipping cut-offs may be the reason, though the data can't say. April 2011 came in 24.42% below April 2010, and I have no evidence for why.

Under a quarter of customers carry the business: 1,376 of 5,824 (23.63%) generate 80% of customer net revenue. The 1,225 Champions alone bring in 68.6% of it. The 1,386 At risk customers hold £1.65M of past revenue, which makes them the obvious win-back list.

The UK gives 85.3% of net revenue and 91.6% of orders, at £459.45 an order. Orders from the other 42 country values average £859.33, 1.87 times as much, led by EIRE (£596,284), Netherlands (£534,369) and Germany (£368,287). The Netherlands averages £2,526 on only 213 orders. The top 20 of 4,720 products make 11.00% of net revenue, with Regency Cakestand 3 Tier (£308,611) first.

The overall return rate is 2.86%, but one cancellation of £77,479.64 on StockCode 23166 (Medium Ceramic Top Storage Jar) in January 2011 pushed that month to 13.62% and makes up 14.2% of all returns. The ten most returned products account for 26.3%.

75.39% of net revenue arrives between 10:00 and 15:59. Thursday is the biggest day (£3.79M), Sunday brings 9.49%, and Saturday almost nothing (0.05%). The forecast puts December 2011 near £773,852 (95% range £672,611 to £875,092), January at £633,547 and February at £554,332, about £1.96M in total. Two years of history is thin, so treat it as a rough baseline.

What I would do

Contact the At risk group before it lapses. Offer European buyers multi-box shipping rates. Log the reason for every credit note, and ask for a deposit on very large single-product orders. Get a customer ID onto guest checkouts, since missing IDs hide 13.3% of net revenue from the customer analysis.

Limitations

(1) The data is a UK retailer's from 2009 to 2011, in pounds. The method would carry over to a Bangladeshi shop but the numbers would not, because Eid, cash on delivery and local delivery costs change the picture. (2) It covers one company, so none of it is an industry benchmark. (3) Each customer sits in the country of their most recent order. 12 of 5,824 bought from more than one country, so country views can be off by a customer or two. Revenue, orders, AOV and return rate are unaffected. (4) Treating any code that doesn't start with a digit as a non-product also catches 220 real lines (gift vouchers, DCGS items, worth £2,808). (5) 11,511 identical lines (0.29% of net revenue) are kept, since a shop can ring up the same item twice. (6) My explanations for monthly patterns are guesses. The data shows only what happened. (7) Built with AI assistance: Gemini made the workbook, and Claude reviewed it and wrote this summary.